LAB SQL Jose Martin

```sql
-- Creación de tabla de Autores
CREATE TABLE Authors (
    author_id INT PRIMARY KEY AUTO_INCREMENT,
    author_name VARCHAR(100) NOT NULL
);

-- Creación de tabla de Artículos
CREATE TABLE BlogPosts (
    post_id INT PRIMARY KEY AUTO_INCREMENT,
    author_id INT NOT NULL,
    title VARCHAR(200) NOT NULL,
    word_count INT NOT NULL,
    views INT NOT NULL,
    FOREIGN KEY (author_id) REFERENCES Authors(author_id)
);

-- Inserción de datos de autores
INSERT INTO Authors (author_name) VALUES
    ('Maria Charlotte'),
    ('Juan Perez'),
    ('Gemma Alcocer');

-- Inserción de datos de artículos
INSERT INTO BlogPosts (author_id, title, word_count, views) VALUES
    (1, 'Best Paint Colors', 814, 14),
    (2, 'Small Space Decorating Tips', 1146, 221),
    (1, 'Hot Accessories', 986, 105),
    (1, 'Mixing Textures', 765, 22),
    (2, 'Kitchen Refresh', 1242, 307),
    (1, 'Homemade Art Hacks', 1002, 193),
    (3, 'Refinishing Wood Floors', 1571, 7542);
```

1	15:15:52	CREATE DATABASE lab_sql	1 row(s) affected	0.016 sec
2	15:15:54	USE lab_sql	0 row(s) affected	0.000 sec
3	15:15:58	CREATE TABLE Authors (author_id INT PRIMARY...	0 row(s) affected	0.032 sec
4	15:17:05	CREATE TABLE BlogPosts (post_id INT PRIMAR...	0 row(s) affected	0.047 sec
5	15:17:08	INSERT INTO Authors (author_name) VALUES ('M...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
6	15:17:13	INSERT INTO BlogPosts (author_id, title, word_count,...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.015 sec

```sql
-- Airline Database Normalization

-- Creación de tabla de Estados de Cliente
CREATE TABLE CustomerStatus (
    status_id INT PRIMARY KEY AUTO_INCREMENT,
    status_name VARCHAR(50) NOT NULL UNIQUE
);

-- Creación de tabla de Clientes
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_name VARCHAR(100) NOT NULL,
    status_id INT,
    total_mileage INT NOT NULL,
    FOREIGN KEY (status_id) REFERENCES CustomerStatus(status_id)
);

-- Creación de tabla de Aeronaves
CREATE TABLE Aircraft (
    aircraft_id INT PRIMARY KEY AUTO_INCREMENT,
    aircraft_name VARCHAR(100) NOT NULL,
    total_seats INT NOT NULL
);

-- Creación de tabla de Vuelos
CREATE TABLE Flights (
    flight_id INT PRIMARY KEY AUTO_INCREMENT,
    flight_number VARCHAR(20) NOT NULL,
    aircraft_id INT NOT NULL,
    mileage INT NOT NULL,
    FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id)
);

-- Creación de tabla de Reservas
CREATE TABLE CustomerFlights (
    customer_flight_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT NOT NULL,
    flight_id INT NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
);

-- Inserción de datos de estados de cliente
INSERT INTO CustomerStatus (status_name) VALUES
    ('None'),
    ('Silver'),
    ('Gold');
```

```sql
-- Inserción de datos de aeronaves
INSERT INTO Aircraft (aircraft_name, total_seats) VALUES
    ('Boeing 747', 400),
    ('Airbus A330', 236),
    ('Boeing 777', 264);

-- Inserción de datos de vuelos
INSERT INTO Flights (flight_number, aircraft_id, mileage) VALUES
    ('DL143', 1, 1351),
    ('DL122', 2, 4370),
    ('DL53', 3, 2078),
    ('DL222', 3, 1765),
    ('DL37', 1, 531);

-- Inserción de datos de clientes
INSERT INTO Customers (customer_name, status_id, total_mileage) VALUES
    ('Agustine Riviera', 2, 115235),
    ('Alaina Sepulvida', 1, 6008),
    ('Tom Jones', 3, 205767),
    ('Sam Rio', 1, 2653),
    ('Jessica James', 2, 127656),
    ('Ana Janco', 2, 136773),
    ('Jennifer Cortez', 3, 300582),
    ('Christian Janco', 2, 14642);

-- Inserción de datos de reservas
INSERT INTO CustomerFlights (customer_id, flight_id) VALUES
    (1, 1), -- Agustine on DL143
    (1, 2), -- Agustine on DL122
    (2, 2), -- Alaina on DL122
    (3, 2), -- Tom on DL122
    (3, 3), -- Tom on DL53
    (4, 1), -- Sam on DL143
    (3, 4), -- Tom on DL222
    (5, 1), -- Jessica on DL143
    (6, 4), -- Ana on DL222
    (7, 4), -- Jennifer on DL222
    (5, 2), -- Jessica on DL122
    (4, 5), -- Sam on DL37
    (8, 4); -- Christian on DL222
```

	#	Time	Action	Message	Duration / Fetch
	1	15:20:57	CREATE TABLE CustomerStatus (status_id INT P...	0 row(s) affected	0.063 sec
	2	15:21:02	CREATE TABLE Customers (customer_id INT PRI...	0 row(s) affected	0.032 sec
	3	15:21:05	CREATE TABLE Aircraft (aircraft_id INT PRIMARY...	0 row(s) affected	0.032 sec
	4	15:21:09	CREATE TABLE Flights (flight_id INT PRIMARY K...	Error Code: 1064. You have an error in your SQL synt...	0.016 sec
	5	15:21:26	CREATE TABLE Flights (flight_id INT PRIMARY K...	0 row(s) affected	0.062 sec
	6	15:21:29	CREATE TABLE CustomerFlights (customer_flight_i...	0 row(s) affected	0.063 sec
	7	15:21:32	INSERT INTO CustomerStatus (status_name) VALUE...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
	8	15:21:35	INSERT INTO Aircraft (aircraft_name, total_seats) VA...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
	9	15:21:38	INSERT INTO Flights (flight_number, aircraft_id, milea...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
	10	15:21:40	INSERT INTO Customers (customer_name, status_id, ...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.000 sec
	11	15:21:43	INSERT INTO CustomerFlights (customer_id, flight_id) ...	13 row(s) affected Records: 13 Duplicates: 0 Warnin...	0.015 sec

In the Airline database write the SQL script to get the total number of flights in the database.

-- 1.Número total de vuelos en la base de datos:
SELECT COUNT(*) AS total_flights
FROM Flights;

	total_flights
▶	5

In the Airline database write the SQL script to get the average flight distance.

-- 2.Distancia promedio del vuelo:
SELECT AVG(mileage) AS average_flight_distance
FROM Flights;

	average_flight_distance
▶	2019.0000

In the Airline database write the SQL script to get the average number of seats.

-- 3. Número promedio de asientos:
SELECT AVG(total_seats) AS average_seats
FROM Aircraft;

	average_seats
▶	300.0000

In the Airline database write the SQL script to get the average number of miles flown by customers grouped by status

```
-- 4. Número promedio de millas voladas por los clientes agrupados por estado:
SELECT cs.status_name, AVG(f.mileage) AS average_miles
FROM CustomerFlights cf
JOIN Customers c ON cf.customer_id = c.customer_id
JOIN CustomerStatus cs ON c.status_id = cs.status_id
JOIN Flights f ON cf.flight_id = f.flight_id
GROUP BY cs.status_name;
```

status_name	average_miles
Gold	2494.5000
None	2084.0000
Silver	2495.3333

In the Airline database write the SQL script to get the maximum number of miles flown by customers grouped by status.

```
-- 5. Número máximo de millas voladas por los clientes agrupados por estado:
SELECT cs.status_name, MAX(c.total_mileage) AS maximum_miles
FROM Customers c
JOIN CustomerStatus cs ON c.status_id = cs.status_id
GROUP BY cs.status_name;
```

status_name	maximum_miles
Gold	300582
None	6008
Silver	136773

In the Airline database write the SQL script to get the total number of aircraft with a name containing Boeing.

```
-- 6. Número total de aeronaves con un nombre que contenga Boeing:
SELECT COUNT(*) AS boeing_aircraft_count
FROM Aircraft
WHERE aircraft_name LIKE '%Boeing%';
```

boeing_aircraft_count
2

In the Airline database write the SQL script to find all flights with a distance between 300 and 2000 miles.

```
-- 7. Vuelos con una distancia entre 300 y 2000 millas:
SELECT *
FROM Flights
WHERE mileage BETWEEN 300 AND 2000;
```

	flight_id	flight_number	aircraft_id	mileage
▶	1	DL143	1	1351
	4	DL222	3	1765
	5	DL37	1	531
*	NULL	NULL	NULL	NULL

In the Airline database write the SQL script to find the average flight distance booked grouped by customer status (this should require a join).

```
-- 8. Distancia promedio del vuelo reservado agrupado por estado del cliente:
SELECT cs.status_name, AVG(f.mileage) AS average_booked_distance
FROM CustomerFlights cf
JOIN Customers c ON cf.customer_id = c.customer_id
JOIN CustomerStatus cs ON c.status_id = cs.status_id
JOIN Flights f ON cf.flight_id = f.flight_id
GROUP BY cs.status_name;
```

	status_name	average_booked_distance
▶	Gold	2494.5000
	None	2084.0000
	Silver	2495.3333

In the Airline database write the SQL script to find the most often booked aircraft by gold status members (this should require a join).

```
-- 9. Aviones reservados con mayor frecuencia por los miembros de estado Gold:
SELECT a.aircraft_name, COUNT(*) AS booking_count
FROM CustomerFlights cf
JOIN Customers c ON cf.customer_id = c.customer_id
JOIN CustomerStatus cs ON c.status_id = cs.status_id
JOIN Flights f ON cf.flight_id = f.flight_id
JOIN Aircraft a ON f.aircraft_id = a.aircraft_id
WHERE cs.status_name = 'Gold'
GROUP BY a.aircraft_name
ORDER BY booking_count DESC
LIMIT 1;
```

	aircraft_name	booking_count
▶	Boeing 777	3